

ONBOARD360 — Customer Onboarding & KYC Process Modernization

Executive Report & Transformation Blueprint | Lead Contributor: Hriday Singh Sobti | NovaBank International

1. Executive Summary: The Strategic Mandate

NovaBank International processes **520,000 onboarding applications annually** across UK/EU, North America, APAC, and Latin America. The legacy journey suffered from an average cycle time of **58.70 hours**, driven by an **89.3% idle queue wait delay (52.42 hours)**. A **33.35% rework defect rate** directly precipitated a **16.25% customer abandonment rate** and inflated unit processing costs to **\$67.45 per completed account** (\$27.97M annual direct operating cost).

The **ONBOARD360 Transformation Initiative** modernizes this fragmented journey into an intelligent, event-driven orchestration ecosystem. By coupling real-time computer vision document gating with automated fuzzy watchlist screening and AI-assisted exception triage, ONBOARD360 compresses average cycle times to **< 24.0 hours** (< 15 minutes for 60% straight-through processing), reduces unit processing costs to **\$12.55** (-81.4%), and generates **\$22,092,276 in annual net recurring cash savings** on an initial **\$2.85M investment**, delivering a **3-Year NPV of \$49.50M** (8.5% hurdle rate), an **IRR > 200%**, and a **2.0-month payback period**.

Operational Metric	AS-IS Baseline	TO-BE Target	Net Variance	Classification
Average Turnaround Time (TAT)	58.70 Hours	< 24.0 Hours (<15m STP)	-59.1% Lead Time	Projected
Idle Queue Wait Time Ratio	89.29% (52.42h)	< 25.00% (5.00h)	-90.5% Queue Drop	Projected
Straight-Through Processing (STP)	0.0% (Batch)	60.0% (Automated)	+60.0 pts Automation	Projected
First-Pass Yield (FPY)	58.12% (302K)	≥ 78.00% (405K)	+19.9 pts Yield Gain	Projected
Application Rework Rate	33.35% (173K)	≤ 10.00% (52K)	-70.0% Defect Cut	Projected
Customer Abandonment Rate	16.25% (84.5K)	≤ 7.50% (39.0K)	-53.8% Churn Cut	Projected
Direct Operating Unit Cost	\$67.45 / Account	\$12.55 / Account	-\$54.90 / Account (-81.4%)	Modelled
Annual Direct Operating OPEX	\$27,967,200.60	\$5,874,924.50	+\$22,092,276.10 Savings	Modelled

2. Key Findings & Pareto Defect Analysis (Figures 1-8)

Empirical process mining proved that **89.3% of turnaround time was idle queue wait time (52.42 hours)**, with active analyst touch time accounting for only 6.29 hours (PCE = 10.71%). Furthermore, **83.15% of all rework loops were driven by three preventable upload flaws**: Blurry Images (42.09%), Expired IDs (23.04%), and Address Typos (18.02%).

Defect Failure Mode	Annual Count	% Rework	Cum %	Root Cause & Architectural Remedy
Blurry / Glare Image	72,994	42.09%	42.09%	Client-side OpenCV WebAssembly real-time sharpness gating (FR-001)
Expired Identification	39,964	23.04%	65.13%	Automated Cloud OCR MRZ date expiration validation within 650ms (FR-002)
Address Proof Mismatch	31,249	18.02%	83.15%	Postal code REST bureau address lookup and prefill (FR-003)
Incomplete Form Fields	20,716	11.94%	95.10%	Interactive form validation and Redis cross-device session cache (FR-020)
Name Typo / Mismatch	8,506	4.90%	100.00%	Jaro-Winkler phonetic double-metaphone fuzzy sanctions matching (FR-010)

3. Financial Appraisal & Value Realization (Figures 11 & 12)

Direct operating costs were modeled using Activity-Based Costing (ABC) and discounted cash flows (8.5% hurdle rate). Operations labor falls by \$12.51M, compliance labor by \$5.27M, support costs by \$1.06M, and vendor fees by \$2.91M, generating **\$22,092,276.10 in net annual recurring cash savings** on an initial **\$2.85M investment** (3-Year NPV \$49.50M, IRR > 200%, payback 2.0 months).

Financial Metric	Value	Financial Metric	Value
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Initial Capital Investment (CAPEX)	\$2,850,000.00	Annual Net Recurring Cash Savings	\$22,092,276.10
Hurdle Discount Rate	8.50%	Year 1 Cash Inflow (80% realization)	\$17,673,820.88
3-Year Net Present Value (NPV)	\$49,501,858.44	Capital Payback Period	2.0 Months
Internal Rate of Return (IRR)	> 200.0%	Conservative Scenario NPV (45% STP)	\$36,063,893.83

4. Quality Assurance, Governance & Board Recommendation

Achieved 100% bidirectional traceability (PNT-01..06 → BR-01..10 → FR-01..35 → Stories → UAT). Machine learning governance audited with **zero regulatory leakage** across 100 PEP/AML cases. Master automated test suite executes **160 test assertions with 100% pass rate in 5.0s**.

Executive Recommendation: Immediate approval of the \$2.85M Phase 1 capital allocation is recommended. Controlled regional pilot cutover commences Month 9 in UK/EU Retail Channel, followed by global rollout in Month 12.